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Efficient simulation of kinetics of radiation induced defects: A cluster dynamics approach

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ABSTRACT

Cluster dynamics has proved to be a useful technique to simulate the microstructure evolution under irradiation over long time scales, especially when rate equations are solved in a deterministic way. However, when the number of equations is large, which is the case when solute atoms are present in addition to self-defects, the numerical solving can become difficult. We show how it is possible to seamlessly couple rate equations to the Fokker–Planck equation, and to increase significantly the efficiency of the numerical algorithms. We apply these methods to the case of helium implantation in α-iron, followed by annealing at high temperature to analyze the effect of surfaces on the coarsening of bubbles.

1. Introduction

The modeling of microstructure evolution of materials under irradiation requires methods which are able to simulate efficiently long time scale phenomena, such as the formation of precipitates, the nucleation, growth and coarsening of bubbles or dislocation loops. These phenomena are the result of elementary events which occur on very short time scales, and which should be properly described to reproduce the final microstructure. One striking example of such phenomena is the constant production and evolution of defects in displacement cascades, whose typical time scales span several orders of magnitude [1], from pico-seconds for the initial development of the cascade to seconds for the long range diffusion of defects.

Accurate modeling techniques such as atomic scale kinetic Monte Carlo (AKMC) are unable to reach large irradiation doses, partly because all atoms in the system are modeled. Some coarse grained methods, such as Object kinetic Monte Carlo (OKMC) [2,3], only retain defects and are therefore more efficient. Even in this case, the fact that some phenomena occur on very short time scales drastically reduces the efficiency of the method, as the time steps used are small. If the modeling is further approximated by using only defect concentrations, the problem can be formulated as a system of ordinary differential equations (ODEs). This is the formalism used in Cluster Dynamics (CD), which simulates the evolution of all cluster types, using one rate equation per cluster type. Though more efficient than OKMC, the stochastic solving of rate equations [4,5] does not bring a tremendous improvement in computation time, as the time scale is still imposed by the most frequent events. The deterministic solving is therefore of particular interest. The fact that some events occur much more frequently than others can however be one of the causes of stiffness for the ODE systems, which render them difficult to be solved by explicit methods. To avoid too small integration time steps, implicit methods are thus necessary, which require the solving of systems of linear equations. For physical systems containing a large number of cluster types, these linear systems can become large, so the computation time and the memory requirements can render simulations unfeasible.

Two ways can be envisaged to reduce the computational burden due to the numerical solving of CD equations. The number of equations to solve can be reduced, by approximating rate equations as a Fokker–Planck equation [6] or by grouping rate equations into classes [7]. Such approximations permit to solve complex problems but still require long simulation times when, for example, a spatial dimension is introduced. An additional way to improve the efficiency is to take advantage of the sparsity of the matrices involved in the linear systems. In this article we present examples of both methods which have been implemented in the CD code CRESCENDO.[1] After presenting shortly the physical models contained in this code, we show that it is possible to couple rigorously rate equations to the discretization of Fokker–Planck equation, and to dramatically speed up the computation. As an example, we then use CRESCENDO to simulate the implantation of

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[1] The acronym CRESCENDO comes from its abbreviation CRESC., which stands for Chemical Rate Equations for the Simulation of Clustering.

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helium in α-iron at a fluence of 10^{16} cm^{-2}, followed by the annealing at 1073 K in two cases, with a thin foil geometry and in the bulk. The effect of surfaces on the coarsening of bubbles produced by helium implantation is discussed.

2. General description of CRESCENDO

CRESCENDO is a CD code which aims at simulating the evolution of clusters up to large doses, due to various phenomena such as irradiation, reactions between clusters and interactions with sinks (mostly dislocations and grain boundaries). CD simulations consist in solving a system of differential equations on cluster concentrations C_ν, where ν refers to the cluster type. A cluster type (or class) is generally defined by the number of elementary components which constitute the cluster, such as interstitials, vacancies, and solute or gas atoms.

The code is made up of different models, which share a common generic part. The generic part contains notably the treatment of the spatial dependency of rate equations (see Section 2.1). For the moment the code contains two models: the simple rate theory model by Sizmann [8] that describes the evolution of mono-vacancies and mono-interstitials only and a more involved model, called IVS model, which is described in Section 2.2 and which will be used in the rest of the article.

2.1. Spatial dependency of equations

In the simplest case, CD equations are ODEs, which implies that the microstructure of the simulated material is homogeneous. It is sometimes necessary to relax this assumption, for example when the simulated system is a thin foil. Effects of surfaces on the spatial distribution of clusters can be so important that a one-dimensional spatial dependency of the rate equations must be introduced. It should be noted that the variation of cluster concentrations should not be too steep, so that the system can still be considered locally homogeneous and treated within a mean-field formalism. The partial differential equations (PDEs) to solve over the foil thickness w_{foil} are thus

$$\frac{\partial C_\mu(z,t)}{\partial t} = R_{\mathrm{model},\mu}(z,\tilde{C}(z,t)) - \frac{\partial J_{\mathrm{d}}(z,t)}{\partial z} = R_{\mathrm{model},\mu}(z,\tilde{C}(z,t)) + \frac{\partial}{\partial z}\left(D_\mu(z)\frac{\partial C_\mu(z,t)}{\partial z}\right), \quad (1)$$

where C_μ is the concentration of mobile species μ which is part of the vector of concentrations $\tilde{C}$, D_μ is the local diffusion coefficient, J_{d} is the diffusion flux and $R_{\mathrm{model},\mu}$ is the reactive part of the equations which is specific to a given model (see Section 2.2).

To solve this PDE system, the spatial operator is discretized on a nonuniform grid. Grid points are indexed from $k = 1$ to $k = N_{\mathrm{grid}}$ and the distance between two grid points z_k and z_{k+1} is h_k (Fig. 1), so for a given species the equation reads (the dependency of C_μ and J_{d} on t has been dropped)

$$\frac{\mathrm{d}C_{\mu,k}}{\mathrm{d}t} = R_{\mathrm{model},\mu,k}(\tilde{C}_k) - 2\frac{J_{\mathrm{d},k+1/2} - J_{\mathrm{d},k-1/2}}{h_{k-1}+h_k}, \quad (2)$$

with

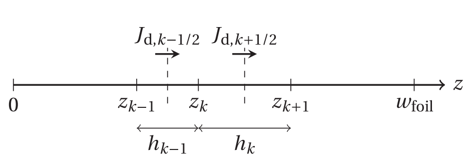

Fig. 1. Schematics of the nonuniform spatial grid. Notations are explained in the text.

$$J_{\mathrm{d},k+1/2} = -D_{\mu,k+1/2}\frac{C_{\mu,k+1}-C_{\mu,k}}{h_k} = -\frac{1}{2}(D_{\mu,k+1}+D_{\mu,k})\frac{C_{\mu,k+1}-C_{\mu,k}}{h_k} \quad (3)$$

and

$$J_{\mathrm{d},k-1/2} = J_{\mathrm{d},(k-1)+1/2} \quad (4)$$

to ensure the continuity of the flux. Concentrations are thus now indexed by (μ, k) and the discretized PDE can be cast into the following ODE:

$$\frac{\mathrm{d}C_{\mu,k}}{\mathrm{d}t} = R_{\mathrm{model},\mu,k}(\tilde{C}_k) + \frac{2}{h_{k-1}+h_k}\left(D_{\mu,k+1/2}\frac{C_{\mu,k+1}-C_{\mu,k}}{h_k} - D_{\mu,k-1/2}\frac{C_{\mu,k}-C_{\mu,k-1}}{h_{k-1}}\right). \quad (5)$$

Similarly, due to the diffusion of solutes along dislocations and grain boundaries, the concentration of solutes in such sinks can be coupled with concentrations in adjacent slices through Fick's law.

Two types of boundary conditions can be used independently on both sides of the system:

- Concentration equal to the thermal equilibrium concentration for self-defects and self-defect clusters, and zero concentration for solutes and clusters containing solutes (Dirichlet condition). This corresponds to the case of a surface acting as a perfect sink and source for defects.
- Zero flux J_{d} on the boundary (Neumann condition). It can correspond for example to the case where an oxide layer at the surface of a metallic sample acts as a barrier for the diffusion of defects.

2.2. IVS model

In this part we focus on the reactive part of the IVS model in a given slice of the system. For the sake of clarity, any dependence of variables on space is omitted in this section. The IVS model can be used if clusters contain interstitials (I) or vacancies (V) and a single type of solute (S). Each cluster type is thus identified by the couple $\nu = (n, p)$, where $|n|$ is the number of interstitials (if $n > 0$) or the number of vacancies (if $n < 0$), and p ($p > 0$) is the number of solutes in this cluster. Examples in α-iron for which CRESCENDO has been used with this model include the simulation of resistivity recovery in α-iron with carbon as an impurity [9], copper precipitation under thermal aging [10] and the simulation of helium implantation [11]. Each of these cases constitutes a submodel of the IVS model. In addition it is possible to duplicate cluster populations containing only self-defects and to give them different properties. This feature is explained later in Section 2.2.3.

2.2.1. Rate equations for clusters with one cluster population

All IVS submodels share the same type of rate equations. For an immobile cluster, the rate equation is

$$\frac{\mathrm{d}C_\nu}{\mathrm{d}t} = \sum_{\mu\in\mathcal{M}_\nu} J_{\nu-\mu,\nu} - \sum_{\mu\in\mathcal{M}} J_{\nu,\nu+\mu} + G_\nu. \quad (6)$$

In this equation, G_ν is the source term due to irradiation and $J_{\nu,\nu+\mu}$ is a net reactive flux between cluster ν and cluster $\nu+\mu$, due to the mobility of cluster μ. It is defined by

$$J_{\nu,\nu+\mu} = \beta_{\nu,\mu}C_\nu C_\mu - \alpha_{\nu+\mu,\mu}C_{\nu+\mu}. \quad (7)$$

The set $\mathcal{M}$ contains all mobile species and $\mathcal{M}_\nu$ is a subset of $\mathcal{M}$, to ensure that the species involved in the flux $J_{\nu-\mu,\nu}$ are well defined in the simulation (for example the number of solutes of $\nu-\mu$ must be non-negative).

The absorption rate $\beta_{\nu,\mu}$ is obtained from the continuous law of diffusion in steady state,

$$\beta_{\nu,\mu} = 4\pi D_\mu Z_{\nu,\mu}(r_\nu + r_\mu + d_{\nu,\mu}), \tag{8}$$

where D_μ and r_μ are respectively the diffusion coefficient and effective radius of cluster μ, $d_{\nu,\mu}$ is a reaction distance and $Z_{\nu,\mu}$ an efficiency factor which accounts for the geometrical shape of the clusters and for elastic effects. Various expressions can be used, depending on the type of cluster and the possible anisotropic diffusion of mobile species for example [12,13].

The emission rate $\alpha_{\nu+\mu,\mu}$ is deduced from the condition $J_{\nu,\nu+\mu} = 0$ at equilibrium, so

$$\alpha_{\nu+\mu,\mu} = \frac{\beta_{\nu,\mu}}{V_{\mathrm{at}}} \exp\left(-\frac{F^{\mathrm{b}}_{\nu+\mu,\mu}}{k_{\mathrm{B}}T}\right), \tag{9}$$

where $F^{\mathrm{b}}_{\nu+\mu,\mu}$ is the binding free energy of cluster μ to cluster ν, which notably includes a configurational entropy contribution, V_{at} the atomic volume, k_{B} the Boltzmann constant and T the temperature.

Although binding free energies can be determined from atomistic computations for clusters containing up to a few hundreds elements [14,10], analytical expressions are needed for larger clusters. For spherical voids and bubbles, a capillary approximation is commonly used [15], possibly modified by the presence of gas inside the cavity [16]. Without gas we can represent a cluster by its self-defect content and note $\nu = n$ and $\mu = m$. In this case the effect of temperature is often neglected and the expression for the emission of a single vacancy is ($n < 0$)

$$F^{\mathrm{b}}_{n,-1} = E^{\mathrm{b}}_{n,-1} = E^{\mathrm{f}}_{-1} + \frac{E^{\mathrm{b}}_{-2,-1} - E^{\mathrm{f}}_{-1}}{2^{2/3}-1}\left(|n|^{2/3} - |n+1|^{2/3}\right). \tag{10}$$

For sufficiently large clusters, the binding energy becomes

$$E^{\mathrm{b}}_{n,-1} = E^{\mathrm{f}}_{-1} - \frac{2\gamma_{\mathrm{s}} V_{\mathrm{at}}}{r_n}, \tag{11}$$

where the surface energy γ_{s} is given by

$$\gamma_{\mathrm{s}} = \frac{E^{\mathrm{f}}_{-1} - E^{\mathrm{b}}_{-2,-1}}{2^{2/3}-1} \frac{1}{\left(36\pi V_{\mathrm{at}}^2\right)^{1/3}}. \tag{12}$$

It should be noted that we could directly use the surface energy given by experiments or atomistic calculations instead of resorting to E^{f}_{-1} and $E^{\mathrm{b}}_{-2,-1}$. The advantage of using Eq. (10) is that it ensures that the binding energy of a di-vacancy is correct, and that binding energies smoothly evolve towards the formation energy of the vacancy for large cluster sizes. In addition, this approximation has been shown to give a correct estimate of the surface energy in α-iron [16].

If clusters are circular dislocation loops and if the material can be treated in the framework of isotropic linear elasticity, the formation energy is given by [17]

$$E^{\mathrm{f}}_n = 2\pi r_n \frac{\mu_{\mathrm{s}} b_\perp^2}{4\pi(1-\nu_{\mathrm{p}})}\left[\ln\left(\frac{4r_n}{\rho_c}\right) - 1\right] + 2\pi r_n \times \frac{\mu_{\mathrm{s}} b_\parallel^2 (2-\nu_{\mathrm{p}})}{8\pi(1-\nu_{\mathrm{p}})}\left[\ln\left(\frac{4r_n}{\rho_c}\right) - 2\right] + \pi r_n^2 \gamma_{\mathrm{sf}}, \tag{13}$$

where μ_{s} is the shear modulus, ν_{p} is the Poisson's ratio, $b_\perp$ is the component of the Burgers vector normal to the plane of the loop, $b_\parallel$ is the component in the plane of the loop, γ_{sf} is the stacking fault energy and ρ_c is a cut-off parameter that can be used as a fitting parameter on atomistic calculations, to notably account for the fact that core contributions are not included in this formula.

The binding energy of a mono-interstitial for example can be deduced from the formation energy of loops, and for large loops it reads

$$E^{\mathrm{b}}_{n,1} = E^{\mathrm{f}}_1 - \frac{V_{\mathrm{at}}\mu_{\mathrm{s}} b_\perp^2}{4\pi(1-\nu_{\mathrm{p}})b_\perp r_n}\ln\left(\frac{4r_n}{\rho_c}\right) - \frac{V_{\mathrm{at}}\mu_{\mathrm{s}} b_\parallel^2(2-\nu_{\mathrm{p}})}{8\pi(1-\nu_{\mathrm{p}})b_\perp r_n}\left[\ln\left(\frac{4r_n}{\rho_c}\right) - 1\right] - \frac{V_{\mathrm{at}}\gamma_{\mathrm{sf}}}{b_\perp}. \tag{14}$$

In CRESCENDO it is implicitly assumed that mobile clusters can only be sufficiently small clusters. More precisely, if $\nu = (n, p)$ is such that $-m_{\mathrm{v}} \leqslant n \leqslant m_{\mathrm{i}}$ and $p \leqslant m_{\mathrm{s}}$, clusters of this type can be mobile. Two additional terms are present and the rate equation reads

$$\frac{\mathrm{d}C_\nu}{\mathrm{d}t} = \sum_{\mu\in\mathcal{M}_\nu} J_{\nu-\mu,\nu} - \sum_{\mu\in\mathcal{M}} J_{\nu,\nu+\mu} - \sum_{\mu\in\Omega} J_{\mu,\nu+\mu} - \sum_i \sum_{j\in\mathcal{N}(i)} k^2_{i,j,\nu} D_\nu (C_\nu - C^{\mathrm{m}}_{i,j,\nu}) + G_\nu. \tag{15}$$

The sum over the whole cluster type set Ω accounts for the absorption and emission of the mobile cluster ν by all clusters. The other term is related to the interaction with sinks. Sinks are identified by their type i (ex.: dislocations, grain boundaries); there are $\mathcal{N}(i)$ sink sub-types of the same type i, which may have different properties (ex.: dislocations with different binding energies for solute atoms). The concentration $C^{\mathrm{m}}_{i,j,\nu}$ is the concentration of species in the matrix close to the sink. For self-defects, $C^{\mathrm{m}}_{i,j,\nu} = C^{\mathrm{eq}}_\nu$ where C^{eq}_ν is the thermal equilibrium concentration. All other concentrations are supposed to be zero near sinks, which also means that they cannot be emitted from sinks, with the exception of single interstitial solute atoms ($\nu = (0,1)$). This is for example the case of helium atoms in various metallic materials. Assuming a local thermodynamic equilibrium near the sink, the chemical potentials of the solute atom in the matrix and in the sink must be the same. For a dislocation, in the dilute limit, we obtain

$$C^{\mathrm{m}}_{\mathrm{d},j,(0,1)} = C_{\mathrm{d},j} K_{\mathrm{d},j}, \tag{16}$$

$$K_{\mathrm{d},j} = \frac{b}{V_{\mathrm{at}}} \exp\left(-\frac{F_{\mathrm{d},j}}{k_{\mathrm{B}}T}\right), \tag{17}$$

where b is the Burgers vector, $C_{\mathrm{d},j}$ is the linear concentration of solutes in the dislocation (in m^{-1}) and $F_{\mathrm{d},j}$ is the binding energy of a solute atom to a dislocation of class j ($j = 1,\ldots,N_{\mathrm{d}}$). Likewise, for grain boundaries we obtain

$$C^{\mathrm{m}}_{\mathrm{gb},j,(0,1)} = C_{\mathrm{gb},j} K_{\mathrm{gb},j}, \tag{18}$$

$$K_{\mathrm{gb},j} = \frac{b^2}{V_{\mathrm{at}}} \exp\left(-\frac{F_{\mathrm{gb},j}}{k_{\mathrm{B}}T}\right), \tag{19}$$

where $C_{\mathrm{gb},j}$ is the surface concentration of solutes in the grain boundary (in m^{-2}) and $F_{\mathrm{gb},j}$ is the binding energy of a solute atom to a grain boundary of class j ($j = 1,\ldots,N_{\mathrm{gb}}$).

2.2.2. Sink strengths and solute concentration in sinks

The expressions for sink strengths are the classical ones, which include the possible anisotropic diffusion of mobile clusters for hcp materials through a parameter $p_{\mathrm{a},\nu}$ defined by

$$p_{\mathrm{a},\nu} = \left(\frac{D_{\nu,c}}{D_{\nu,a}}\right)^{1/6}, \tag{20}$$

where $D_{\nu,c}$ and $D_{\nu,a}$ are the diffusion coefficients along the c- and a-axis of the hcp lattice [18].

Two different types of sinks can be used:

1. *Dislocations*. The sink strength reads

$$k^2_{\mathrm{d},j,\nu} = \rho_{\mathrm{d},j} Z_{\mathrm{d},j,\nu},$$

where $\rho_{d,j}$ is the dislocation density of class j and $Z_{d,j,v}$ is an efficiency factor. In the framework of the effective medium approximation, the efficiency factor depends on the total sink strength [19]. Since such multiple sink effects have not been taken into account in the efficiency factors of clusters, we do not consider them for dislocations. A further usual simplification consists in taking $Z_{d,j,v}$ independent of the dislocation density, so we have finally

$$Z_{d,j,v} = Z_{d,v} \frac{\sqrt{\cos^2 \lambda_{d,j} + p_{a,v}^6 \sin^2 \lambda_{d,j}}}{p_{a,v}^2}, \tag{21}$$

where $Z_{d,v}$ is a constant and $\lambda_{d,j}$ is the angle between the dislocation line and the c-axis. Currently all dislocation densities are constant. The equation on the linear solute concentration in a given class j of dislocations is ($v = (n, p)$)

$$\frac{\mathrm{d}C_{d,j}}{\mathrm{d}t} = \sum_{v \in \mathcal{M}} p Z_{d,j,v} D_v C_v - Z_{d,j,(0,1)} D_{(0,1)} K_{d,j} \, C_{d,j}. \tag{22}$$

2. *Grain boundaries*. In general the multiple sink effects cannot be neglected, which means the sink strength depends on the total sink strength in a single crystal $k_{sc,v}^2$, *i.e.* excluding grain boundaries [20]. When the sink strength of the single crystal is large, the sink strength of the grain boundary is given by

$$k_{gb,j,v}^2 = \frac{6}{\bar{l}_{gb}} x_j k_{sc,v}, \tag{23}$$

where

$$\frac{1}{\bar{l}_{gb}} = \frac{1}{3}\left(\frac{2}{p_{a,v}} + p_{a,v}^2\right)\frac{1}{l_{gb}}. \tag{24}$$

In these equations, l_{gb} is the grain diameter and x_j is the proportion of grains of class $j\left(\sum_{j=1}^{N_{gb}} x_j = 1\right)$. The total sink strength $k_{sc,v}^2$ is defined by:

$$k_{sc,v}^2 = \sum_j k_{d,j,v}^2 + \frac{S_v}{D_v},$$

where

$$S_v = \sum_{\mu \in \Omega} \beta_{\mu,v} C_\mu + \sum_{\mu \in \mathcal{M}} \beta_{v,\mu} C_\mu. \tag{25}$$

To derive the equation on the surface solute concentration $C_{gb,j}$ in a given grain boundary class j, the concentration of grain boundaries of such a class is needed. It is approximately $3x_j/l_{gb}$, so we obtain finally ($v = (n, p)$)

$$\frac{\mathrm{d}C_{gb,j}}{\mathrm{d}t} = \frac{l_{gb}}{3x_j}\left(\sum_{v \in \mathcal{M}} p k_{gb,j,v}^2 D_v C_v - k_{gb,j,(0,1)}^2 D_{(0,1)} K_{gb,j} C_{gb,j}\right). \tag{26}$$

2.2.3. Additional cluster populations

So far we have considered a single population of clusters which can contain interstitials or vacancies and solutes. In CRESCENDO, it is also possible to duplicate as many times as desired a cluster type $(n, 0)$ which contains no solute atom. This is useful when physical properties of two cluster configurations are so different that it is irrelevant to take them into account through the configurational entropy term (Section 2.2.1). This is the case for example when one wants to simulate $\langle 100 \rangle$ and $\langle 111 \rangle$ interstitial loops which are known to coexist at certain temperatures in α-iron [21], $\langle a \rangle$ and $\langle c \rangle$ type vacancy loops in zirconium [22], or stacking fault tetrahedra and voids in austenitic stainless steels [23]. The different types of clusters are assumed to be produced either inside displacement cascades, or through homogeneous nucleation. Mobile clusters are not duplicated, so it is necessary to provide

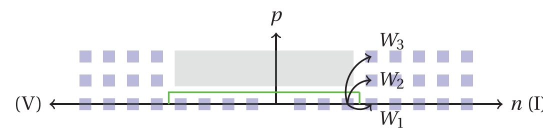

Fig. 2. Schematic representation of the cluster space when three cluster populations are used and no solute is present. Each cluster type is represented by a blue square and identified with a couple (n, p), where $|n|$ is the number of self-defects and p is the population index. The probability that two mobile clusters give a cluster of type i when they react with each other is W_i. The green line surrounds the part where clusters can be mobile. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

the probability W_i that two mobile clusters which react with each other produce a cluster of a given population i (Fig. 2).

3. Simulation of large clusters: Fokker–Planck formulation

Simulating systems of interest by solving directly the ODE system (5) is often unfeasible, even with efficient numerical solvers and processors. Several methods have thus been proposed to reduce the number of equations to solve. Some of them consist in grouping cluster classes and performing exchange of clusters between groups. Such a method was developed by Kiritani [24] and used by Hayns [25]. Later, grouping methods were improved by Golubov et al. [7] and generalized to the case where both self-defects and solute atoms are present [26]. As noticed by Wolfer et al. [6], for sufficiently large clusters rate equations can be well approximated by a Fokker–Planck equation in the cluster space. The discretization of this PDE on a coarse mesh provides an efficient way to determine the cluster kinetics with time. The transformation of the rate equations into a Fokker–Planck equation is also interesting from a physical point of view, since the advection and diffusion terms can be related to the cluster growth and to the cluster size fluctuations. The Fokker–Planck formulation has been used to simulate various problems, such as the clustering of vacancies and interstitials [27], possibly with helium [28] or the precipitation in a binary alloy [29].

Two questions arise when Fokker–Planck formulation is used. The first one is related to the discretization scheme used. It is known that using central differences for the advective term [27,30] can lead to stability problems. The second question is how to couple the discretized Fokker–Planck equation to the rate equations used for small clusters. An original scheme has been developped to that purpose, which ensures the conservation of matter.

3.1. Numerical scheme for the Fokker–Planck equation

For the sake of simplicity we consider in the following a system with only self-defects, so v can be represented by its first index n, and whose source term is set to zero. Negative and positive indexes represent vacancy and interstitial clusters respectively. If we assume that mobile clusters are small clusters, such that $-m_v \leqslant n \leqslant m_i$, Eq. (6) for immobile clusters reads

$$\frac{\mathrm{d}C_n}{\mathrm{d}t} = \sum_{m=-m_v}^{m_i} (\beta_{n-m,m} C_{n-m} C_m - \alpha_{n,m} C_n) - \sum_{m=-m_v}^{m_i} (\beta_{n,m} C_n C_m - \alpha_{n+m,m} C_{n+m}). \tag{27}$$

Concentrations $C_n(t)$ can be viewed as particular values of a scalar field $C(x, t)$ at grid points $x = x_n$. The same reasoning holds for absorption and emission coefficients $\beta_{n,m}$ and $\alpha_{n,m}$, which can be written $\beta_m(x_n)$ and $\alpha_m(x_n)$. Assuming $n \gg m$, a partial differential equation associated to (27) can be obtained by a Taylor expansion up to second order, which gives

$$\frac{\partial C}{\partial t} = -\frac{\partial J}{\partial x}, \tag{28}$$

with

$$J = FC - \frac{\partial DC}{\partial x} \tag{29}$$

and

$$F(x) = \sum_{m=-m_v}^{m_i} m[\beta_m(x)C_m - \alpha_m(x)] \tag{30}$$

$$D(x) = \frac{1}{2}\sum_{m=-m_v}^{m_i} m^2[\beta_m(x)C_m + \alpha_m(x)]. \tag{31}$$

As discussed in the introduction of this section, $F(x)$ is a drift (or advection) term and is related to the growth or shrinkage of a cluster of size x, depending on its sign. On the contrary $D(x)$ is always positive and acts as a diffusion term in the cluster space. It is related to cluster size fluctuations and is notably responsible for cluster nucleation.

The discretization of the Fokker–Planck equation is done by a finite volume approach on a nonuniform grid in the cluster space (Fig. 3). Integrating Eq. (28) over the "volume" of the element n leads to

$$\frac{\mathrm{d}C_n}{\mathrm{d}t} = \frac{2}{\Delta x_n + \Delta x_{n-1}}(J_{n-1/2} - J_{n+1/2}). \tag{32}$$

The expression for $J_{n+1/2}$ results from the discretization of Eq. (29). The diffusion term is discretized by a centered scheme which depends on $(DC)_n$ and $(DC)_{n+1}$. The simplest scheme for the drift term $(FC)_{n+1/2}$ consists in using the average between $(FC)_n$ and $(FC)_{n+1}$, which leads to an overall centered scheme for the PDE. Such a symmetric stencil would be interesting, notably because for a uniform grid of width Δx it is shown to be on the order of $(\Delta x)^2$, whereas a 2-point asymmetric scheme is on the order of Δx. In addition, if $\Delta x = 1$ and if only monomers are mobile, it can be shown that the initial rate equations are recovered [30]. However, it is well-known that for PDEs with strong advection terms, centered schemes for the advection term are unstable [31]. This can be problematic in CD where advection fluxes can be large enough for unstabilities to develop, which results in large oscillations and negative concentrations.

To derive a practical scheme for the flux (29), we follow the approach developed by Chang and Cooper for certain Fokker–Planck equations [32], and adapt it for the present case. The underlying idea is that an accurate scheme should at least properly describe the steady state solution of the Fokker–Planck equation, given by

$$\frac{\partial}{\partial x}\left(FC - \frac{\partial DC}{\partial x}\right) = 0. \tag{33}$$

As we consider large clusters, the expressions for F and D can be simplified as follows. Coefficients α_m and β_m in Eqs. (30) and (31) can be expanded asymptotically as a function of r, using either Eq. (11) or (14) depending on the cluster type considered. In both cases,

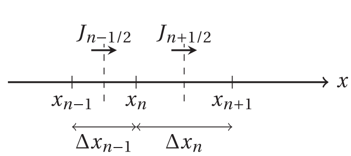

Fig. 3. Schematics of a nonuniform mesh used for the finite-volume evaluation of the Fokker–Planck equation. Only one-dimensional Fokker–Planck equation is envisaged here.

the leading term is proportional to r. If only this term is kept, we find that $DC \approx \gamma FC$, where γ is a constant. If this relation is reported in Eq. (33), we obtain

$$(FC)(x) = A + B\exp\left(\frac{x}{\gamma}\right), \tag{34}$$

where A and B are two constants. They are determined on a discretization interval $[x_n,\ x_{n+1}]$ by imposing that $(FC)(x_n) = (FC)_n$ and $(FC)(x_{n+1}) = (FC)_{n+1}$. On this interval the flux is therefore

$$J_{n+1/2} = \frac{(FC)_{n+1} - (FC)_n\exp(w_n)}{1 - \exp(w_n)}, \tag{35}$$

where $w_n = \Delta x_n/\gamma$. This flux can be also written

$$\begin{aligned} J_{n+1/2} = {} & \left(\frac{1}{1-\exp(w_n)} + \frac{1}{w_n}\right)(FC)_{n+1} - \frac{1}{w_n}(FC)_{n+1} \\ & - \left(\frac{\exp(w_n)}{1-\exp(w_n)} + \frac{1}{w_n}\right)(FC)_n + \frac{1}{w_n}(FC)_n, \end{aligned} \tag{36}$$

$$= (1-\delta_n)(FC)_{n+1} + \delta_n(FC)_n - \frac{(DC)_{n+1} - (DC)_n}{\Delta x_n}, \tag{37}$$

which provides a discretization scheme for Eq. (29). In this expression δ_n is given by

$$\delta_n = -\frac{1}{\exp(-w_n) - 1} - \frac{1}{w_n}. \tag{38}$$

In practice, w_n is calculated on each interval as

$$w_n = \frac{(F_n + F_{n+1})\Delta x_n}{D_n + D_{n+1}}. \tag{39}$$

This quantity gives the relative strengths of advection and diffusion terms. If it is close to zero, $\delta_n \approx 1/2$ and the advective part of the flux is the mean of $(FC)_n$ and $(FC)_{n+1}$. We recover the centered scheme, which is appropriate for purely diffusive processes. If the advection term multiplied by Δx_n is far larger than the diffusion term, δ_n tends towards 1 or 0 depending on the sign of FC and the advective term is simply $(FC)_n$ (if $FC > 0$) or $(FC)_{n+1}$ (if $FC < 0$). We obtain the upwind scheme for the advection term, which is known to be stable.

3.2. Connecting rate equations and the Fokker–Planck equation

An additional difficulty in the discretization scheme is due to the fact that the Fokker–Planck equation is not used over the whole cluster space. For small clusters, rate equations are preserved, so a special scheme is necessary to couple the two formalisms. The part in the cluster space where initial rate equations are used will be called the "discrete part" and the part where Fokker–Planck equation is used will be called the "continuous part". As discussed in the previous section, the coupling between the two parts is straightforward when a centered scheme is used on a mesh with $\Delta x = 1$ and when only monomers are mobile, since the discretized Fokker–Planck equation simplifies to the discrete rate equations. However, as soon as clusters containing several self-defects are mobile, discrete rate equations involve the coupling between non-adjacent classes, whereas Fokker–Planck remains a local approach whatever the value of Δx, coupling only adjacent classes when it is discretized with the scheme exposed previously.

To make the link between the discretized Fokker–Planck equation and the rate equations, it is interesting to write the discretized Fokker–Planck Eq. (32) as a sum of partial fluxes involving the different types of mobile clusters,

$$\frac{\mathrm{d}C_n}{\mathrm{d}t} = \frac{2}{\Delta x_n + \Delta x_{n-1}}\sum_{m=-m_v}^{m_i}\left(J^m_{n-1/2} - J^m_{n+1/2}\right), \tag{40}$$

where

$$J^m_{n+1/2} = (1-\delta_n)(FC)^m_{n+1} + \delta_n (FC)^m_n - \frac{(DC)^m_{n+1} - (DC)^m_n}{\Delta x_n} \quad (41)$$

and

$$(FC)^m_n = m[\beta_{n,m} C_m - \alpha_{n,m}] C_n, \quad (42)$$

$$(DC)^m_n = \frac{1}{2} m^2 [\beta_{n,m} C_m + \alpha_{n,m}] C_n. \quad (43)$$

If $\Delta x = 1$ and if a centered scheme is used ($\delta_n = 1/2$), we have

$$\begin{aligned} J^m_{n-1/2} - J^m_{n+1/2} = & \frac{m(m+1)}{2} \beta_{n-1,m} C_{n-1} C_m + (1-m^2)\beta_{n,m} C_n C_m \\ & + \frac{m(m-1)}{2} \beta_{n+1,m} C_{n+1} C_m - \beta_{n,m} C_n C_m - \alpha_{n,m} C_n \\ & + \frac{m(m-1)}{2} \alpha_{n-1,m} C_{n-1} + (1-m^2)\alpha_{n,m} C_n \\ & + \frac{m(m+1)}{2} \alpha_{n+1,m} C_{n+1}. \end{aligned} \quad (44)$$

In this equation, the first three terms and the last three terms are second order approximations of $\beta_{n-m,m} C_{n-m} C_m$ and $\alpha_{n+m} C_{n+m}$ respectively. Since these terms are present as such in the discrete rate equations, even for centered schemes on a mesh with $\Delta x = 1$ the second-order approximation contained in the Fokker–Planck equation introduces an error which can lead to a loss of matter at the interface between the discrete and the continuous parts. To circumvent this problem, a transition part is introduced between the discrete and the continuous parts (Fig. 4). The number of classes contained in this part is $N_t = \max(m_i, m_v)$, so that clusters in the discrete part are only coupled to the transition part, and not to the continuous part. In this transition part $\Delta x = 1$ and the Fokker–Planck equation is used with a centered scheme. This transition zone is sufficiently thin to avoid any instability caused by the use of the centered scheme. In addition, when terms in Eq. (44) are due to exchanges between the transition and the discrete parts, they are replaced by their exact counterpart if necessary, so the terms from clusters in the discrete part and in the transition part are fully consistent. Outside the transition part, the discretization scheme is not necessarily centered and the mesh step can increase. The fact that $J_{n+1/2} = J_{(n+1)-1/2}$ ensures the conservation of matter.

3.3. Equations for mobile clusters

When Fokker–Planck equation is used, Eq. (15) must be modified to ensure that the absorption and emission fluxes of mobile clusters are consistent with the fluxes present in the Fokker–Planck equation. On a given interval $[x_{i_1}, x_{i_2}]$ in the continuous part, the variation rate of the quantity of matter reads

$$\frac{\mathrm{d}\mathcal{S}}{\mathrm{d}t} = \sum_{n=i_1}^{i_2} x_n \frac{x_{n+1} - x_{n-1}}{2} \frac{\mathrm{d}C_n}{\mathrm{d}t} = \sum_{n=i_1}^{i_2} x_n [J_{n-1/2} - J_{n+1/2}]. \quad (45)$$

From this expression, the contribution of a given mobile cluster m to the variation rate in this interval is

$$\begin{aligned} \frac{\mathrm{d}\mathcal{S}^m}{\mathrm{d}t} &= \sum_{n=i_1}^{i_2} x_n \left[J^m_{n-1/2} - J^m_{n+1/2} \right] \\ &= x_{i_1} J^m_{i_1-1/2} - x_{i_2} J^m_{i_2+1/2} + \sum_{n=i_1+1}^{i_2} \Delta x_{n-1} J^m_{n-1/2}, \end{aligned} \quad (46)$$

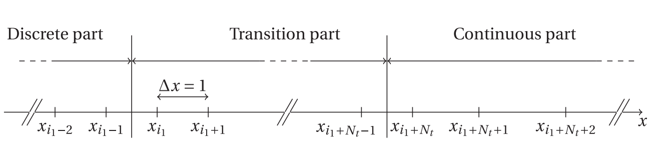

Fig. 4. Schematics of the transition part used to ensure a seamless coupling between the discrete and continuous parts.

where we have used the relation $J^m_{n+1/2} = J^m_{(n+1)-1/2}$. The first and second term in the last expression correspond to the fluxes of clusters which absorb mobile clusters and therefore cross the interfaces at $n = i_1$ and $n = i_2$, while the third term represents the variation of matter due to absorption or emission of mobile clusters by clusters which remain in the interval.

The equation for the mobile cluster m must therefore contain the following term:

$$\frac{\mathrm{d}C_m}{\mathrm{d}t} = -\frac{1}{m} \sum_n \Delta x_{n-1} J^m_{n-1/2} - \cdots \quad (47)$$

The summation indexes must be carefully chosen, since fluxes $J^m_{n-1/2}$ are not used as such in the transition part. The complete expression is derived in Appendix A.

3.4. Numerical validation of the discretization scheme and comparison with grouping methods

To validate the numerical scheme proposed in the previous sections, we consider the simulation of α-iron under neutron irradiation with the parametrization used in Ref. [33], and which is reported with our notations in Table 1 for clarity. In this parametrization, $m_i = 3$ and $m_v = 4$, which offers a good test for the numerical scheme and in particular for the treatment of the coupling between the discrete and continuous parts. The cluster distribution is shown in Fig. 5 at 0.2 dpa ($t = 1.5 \times 10^6$ s), for both vacancy ($n < 0$) and interstitial clusters ($n > 0$). A geometric progression for the discretization is used in the continuous part, *i.e.* $\Delta x_{n+1} = \eta \Delta x_n$. In the case of $\eta = 1.1$, the cluster distribution is simulated with only 231 classes whereas 37,976 classes are needed for a fully discrete simulation.

It can be seen that a very good agreement is obtained between the two simulations, except for the distribution tail in the interstitial part. This discrepancy is due to our numerical scheme which introduces numerical diffusion through the discretization of the advection term. Indeed, as shown in Fig. 5(d), δ_n is close to one for large interstitial clusters, so that the numerical scheme (37) is simply an upwind scheme for the advective term. It is known that the upwind scheme, though stable, introduces substantial numerical diffusion [31]. It can be shown (see Appendix B) that provided $DC = \gamma\, FC$, which is true for large interstitial clusters (Fig. 5(b)), the physical diffusion term is multiplied by

$$f_n = \frac{\Delta x^2_{n-1}}{\Delta x_{n-1} + \Delta x_n} \frac{1}{\gamma} = \frac{\Delta x_{n-1}}{1+\eta} \frac{1}{\gamma}$$

for a geometrical mesh.

Although $F_n/D_n = 1/\gamma$ is small (Fig. 5(b)), $w_n = \Delta x_n/\gamma$ can be very large due to the geometrical progression of Δx_n (Fig. 5(c)), which leads to the numerical diffusion observed. By reducing the geometrical progression η, the numerical diffusion is reduced without too much increase in computational time. For $\eta = 1.03$ (Fig. 5(a)), a better agreement is obtained, and 455 classes are needed to reproduce the cluster distribution, which remains far below the number of classes in the discrete case.

To evaluate the accuracy of the numerical scheme, it is also useful to check that the matter is conserved between two times t_1 and t_2. The following quantity is calculated regarding interstitials and vacancies:

$$\begin{aligned} \Delta S_{\mathrm{iv}} = & \sum_n \frac{x_{n+1} - x_{n-1}}{2} x_n (C_n(t_2) - C_n(t_1)) \\ & - \int_{t_1}^{t_2} \sum_i \sum_{j \in \mathcal{N}(i)} k^2_{i,j,n} x_n D_n \left(C_n(t) - C^{\mathrm{m}}_{i,j,n} \right) \mathrm{d}t \\ & - \int_{t_1}^{t_2} \sum_n x_n G_n(t) \mathrm{d}t. \end{aligned} \quad (48)$$

Table 1
Parameters used for the simulation of the test case described in Ref. [33]. The recombination radius r_{iv} is the minimum interaction distance between interstitial and vacancy clusters. Absorption efficiencies by loops and cavities are described in Ref. [34], and are just an extension for mobile clusters containing several self-defects of those used in Ref. [12] for single self-defects.

Symbol	Description	Value	Unit
V_{at}	Atomic volume	1.182×10^{-23}	cm^{-3}
m_i	Index of the largest mobile interstitial cluster	3	
m_v	Index of the largest mobile vacancy cluster	4	
D_ν^0	Diffusion prefactor of clusters	8.2×10^{-3}	$cm^2\,s^{-1}$
E_1^m, E_2^m, E_3^m	Migration energy of interstitial clusters	0.34, 0.42, 0.43	eV
$E_{-1}^m, E_{-2}^m, E_{-3}^m, E_{-4}^m$	Migration energy of vacancy clusters	0.83, 0.62, 0.35, 0.48	eV
E_1^f	Formation energy of the mono-interstitial	3.64	eV
E_{-1}^f	Formation energy of the mono-vacancy	2.2	eV
$E_{2,1}^b, E_{3,1}^b, E_{4,1}^b$	Binding energy for interstitial clusters	0.83, 0.92, 1.64	eV
$E_{-2,-1}^b, E_{-3,-1}^b, E_{-4,-1}^b, E_{-5,-1}^b$	Binding energy for vacancy clusters	0.30, 0.37, 0.62, 0.73	eV
$E_{\nu,1}^b (\nu > 4)$	Binding energy for interstitial clusters	Capillary law	
$E_{\nu,-1}^b (\nu < -5)$	Binding energy for vacancy clusters	Capillary law (Eq. (10))	
$Z_{d,\mu}$ $(\mu > 0)$	Absorption efficiency (straight dislocations)	1.1	
$Z_{d,\mu}$ $(\mu < 0)$	Absorption efficiency (straight dislocations)	1.	
r_{iv}	Recombination radius	0.65	nm
l_{gb}	Grain size	200	µm
ρ_d	Dislocation density	10^8	cm^{-2}
T	Temperature	573	K
G_{-1}, G_{-8}	Creation rate of vacancy clusters	7.8×10^{-9}, 3.9×10^{-9}	dpa s^{-1}
G_1, G_4	Creation rate of interstitial clusters	3.9×10^{-8}, 2.9×10^{-12}	dpa s^{-1}

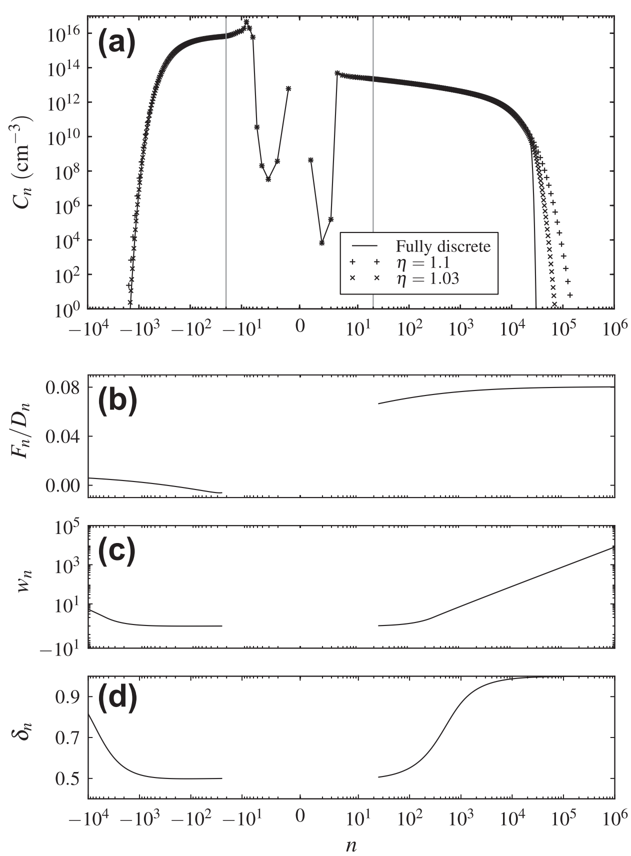

Fig. 5. (a) Cluster distribution C_n calculated at 0.2 dpa using only discrete rate equations and with Fokker–Planck formalism in the case of two different geometric progressions ($\Delta x_{n+1} = \eta \Delta x_n$ with $\eta = 1.1$ or 1.03). The discrete part is bounded by the two vertical lines. (b–d) The three quantities are related to the Fokker–Planck equation in the continuous part and are discussed in the text.

For spatialized simulations, the flux between slices should also be included in the formula. When solutes or additional cluster populations are present, expressions are slightly more complex since summation should also be performed on a second index p. A similar equation can also be written for the conservation of solute atoms.

The conservation can then be assessed by calculating

$$\epsilon_{iv} = \frac{\Delta S_{iv}}{\max(S(t_1), S(t_2))}, \tag{49}$$

where

$$S(t) = \max(S_i(t), S_v(t))$$

and (θ = i, v)

$$S_\theta(t) = \sum_n \frac{x_{n+1} - x_{n-1}}{2} |x_n| C_n(t).$$

The quantity ϵ_{iv} is represented in Fig. 6 as a function of time for the same test case as before. Quantities $S^d = S_i^d + S_v^d$ and $S^c = S_i^c + S_v^c$ are also represented, where S_θ^d (S_θ^c) is the restriction of S_θ to the discrete (continuous) part. Such a partitioning permits to see when the number of self-defects in the continuous part is larger than in the discrete part. It can be seen that no significant variation of ϵ_{iv} occurs when the contribution of the continuum becomes very large, and that it remains close to the machine precision.

It is interesting to see what is the influence of numerical diffusion on the cluster distribution when its exhibits a maximum in the continuous part, in contrast with the previous test. Such a case has been treated by Ovcharenko et al. [35] to show the accuracy of their grouping method. Starting with an atomic fraction of mono-vacancies equal to 10^{-7} in a nickel-like metal, the authors simulated the evolution of the cluster distribution at 823 K. We show in Fig. 7 the cluster distribution $C(r) = C_n dn/dr$ obtained at $t = 10^5$ s and $t = 10^8$ s considering all classes and with our numerical scheme, using a discrete part of 10 classes and then a continuous part containing 500 classes with a geometrical progression equal to $\eta = 1.03$. We also show the result obtained by a centered difference scheme ($\delta_n = 0.5$), which turned out to produce no oscillations in the cluster distribution. It can be seen that the centered scheme satisfactorily reproduces the cluster distribution at both times, whereas the Chang–Cooper scheme, due to numerical diffusion, induces a substantial broadening of the peak. The mean radii

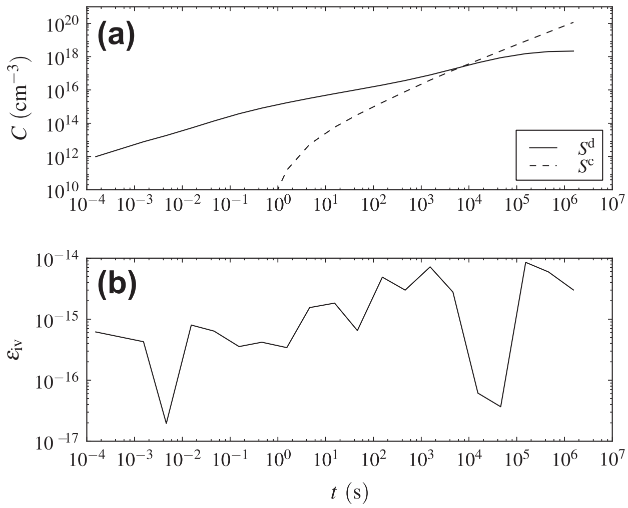

Fig. 6. (a) Concentration of self-defects contained in the discrete and continuous parts in the case of neutron irradiation of α-iron. (b) Relative error on the matter conservation.

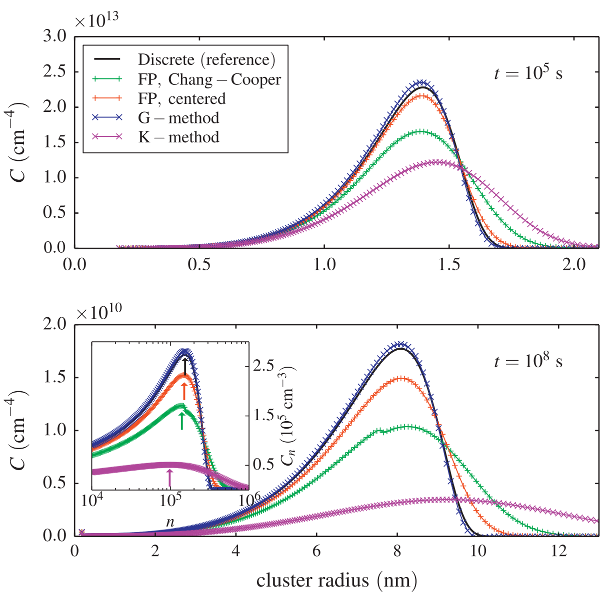

Fig. 7. Cluster distribution at $t = 10^5$ s and $t = 10^8$ s for the test case proposed in Ref. [35], using two different schemes (Chang–Cooper and centered) for the Fokker–Planck approach, the Golubov method described in Ref. [7] (G-method) and the Kiritani method [24] (K-method). The inset at $t = 10^8$ s shows the distribution as a function of n, to exhibit the position of the peak (it is highlighted with an arrow).

and densities obtained at $t = 10^5$ s are 1.3 nm and 1.03×10^{13} cm^{-3} for the fully discrete case, 1.3 nm and 1.02×10^{13} cm^{-3} for the centered scheme, and 1.3 nm and 9.40×10^{13} cm^{-3} for the Chang–Cooper scheme. At $t = 10^8$ s, the mean radii and the densities for the three methods are respectively 7.2 nm and 5.67×10^{10} cm^{-3}, 7.4 nm and 5.30×10^{10} cm^{-3}, and 7.7 nm and 4.56×10^{10} cm^{-3}. It can thus be concluded that for both schemes the results are acceptable even at large times, regarding the experimental uncertainties. It should be noted that more generally, in all cases investigated, we have checked that the mean radius and the density of clusters are well reproduced with the Fokker–Planck approach, whatever the discretization scheme. The relative error on the conservation of matter as computed by Eq. (49) is less than 10^{-12} in all cases.

Our solution can be compared to grouping methods, which do not rely on the discretization of the Fokker–Planck equation. For this comparison we chose the following two grouping methods: the Kiritani method [24] (K-method) and the grouping method proposed by Golubov et al. [7] (G-method), which uses two equations per cluster class to ensure the conservation of cluster density and the conservation of matter. We introduced these methods as modules of CRESCENDO, following the implementation provided in Ref. [35] for the G-method, which enabled us to solve the equations with CVODES (see Section 4) and to use the same discretization in cluster space for all methods. It can be seen in Fig. 7 that the G-method reproduces remarkably well the exact solution, except in the distribution tail where spurious oscillations with large negative concentrations are present (Fig. 8). Such oscillations can be also observed with the Modified Euler method implemented in Ref. [35]. Concerning the K-method, the distribution broadens more than with the other schemes and it also appears shifted towards large sizes: the mean radius and the density are 1.39 nm and 8.12×10^{12} cm^{-3} at $t = 10^5$ s, and 9.2 nm and 2.44×10^{10} cm^{-3} at $t = 10^8$ s. It should be noted that the shift in the distribution does not come from a too large advection term. In fact, as derived in Appendix B and in agreement with Ref. [7], the distribution C_n is shifted towards the small values of n with respect to the exact solution (see the inset of Fig. 7 at $t = 10^8$ s). However, since the distribution $C(r)$ involves a multiplication of C_n by r^2 and since the peak is significantly broader in the case of the K-method, an artificial shift towards large radii is obtained for $C(r)$. Such a shift is also seen with the discretization of the Fokker–Planck equation due to numerical diffusion, but to a much lesser extent.

The use of the same ODE solver for the different methods permits a direct comparison of the performances of the methods. To perform this test, a finite-difference computation of the jacobian matrix was used for the grouping methods and for the Fokker–Planck scheme. The computation time to achieve a physical time equal to $t = 10^8$ s is 27 s for the K-method, 80 s for the Fokker–Planck method and 39 min for the G-method on a 3.07 GHz processor. Actually the computation of the jacobian matrix takes some time in the computation process, and since the G-method uses twice as many equations as the Fokker–Planck scheme, it may affect the performance. Therefore we also carried out a computation with the Fokker–Planck scheme, using 1000 equations in the continuous part instead of 500. The computation time is 4 min in this case, still far below the time needed to complete the calculation with the G-method and 500 classes in the continuous part. The reason is that during the integration process, the integration time step

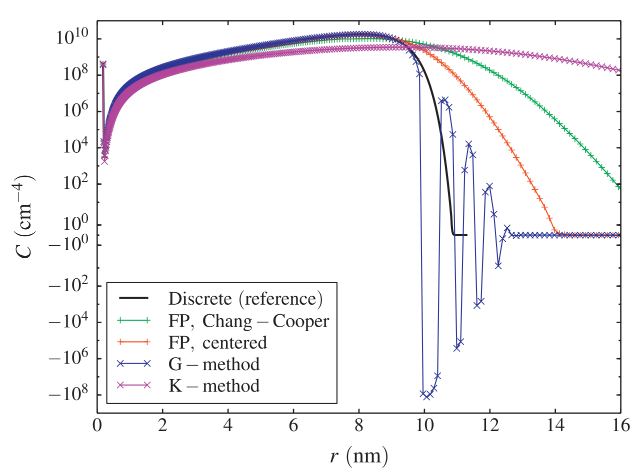

Fig. 8. Cluster distributions at $t = 10^8$ s for the test case proposed in Ref. [35], using two different schemes (Chang–Cooper and centered) for the Fokker–Planck approach, the Golubov method described in Ref. [7] (G-method) and the Kiritani method [24] (K-method). The scale for C is linear between -1 and 1 and logarithmic outside this range.

Δt is in general smaller with the G-method and frequent updates of the jacobian matrix are needed (Fig. 9). Although it is not clear to us what causes this difficulty for CVODES to integrate grouped equations, it may be linked to the appearance of negative concentrations with oscillations in the cluster distribution tail. The oscillations in the cluster distribution and the reduction of the integration time step were not seen when only zeroth order moments were taken into account, but in this case the conservation of matter could not be guaranteed.

To conclude, our Fokker–Planck approach based on a centered scheme should be used whenever it leads to non-oscillating solutions, since it is more accurate than the Chang–Cooper scheme. The Chang–Cooper scheme is stable and it is useful in all cases to get mean radii and cluster densities, which is often satisfactory when results are compared to experimental data, but it may introduce too much numerical diffusion when cluster distributions are peaked. To avoid such diffusion, higher order discretization schemes could be envisaged (see for example Ref. [36] and the various schemes based on this method), but it seems to us that in most cases, the Chang–Cooper scheme provides enough precision for the expected applications. The G-method by Golubov et al., though slower than the Fokker–Planck approaches, reproduces closely the exact distribution at all times for the test case presented here, except for the distribution tail where oscillations with negative concentrations are present. The distributions obtained by the K-method are significantly shifted with respect to the reference calculation, contrary to the other methods tested.

3.5. Extension of the numerical scheme to higher dimensions

In the previous sections no solute was considered, which resulted in the discretization of a one-dimensional Fokker–Planck equation. Similar one-dimensional PDEs exist when solute atoms are present and when the Taylor expansion is carried out only along the self-defect axis. In the case of a single additional type of solute, the PDE is

$$\frac{\partial C_{(.,p)}}{\partial t} = R_{(.,p)}(x) - \frac{\partial}{\partial x}\left(FC_{(.,p)} - \frac{\partial DC_{(.,p)}}{\partial x}\right), \tag{50}$$

where a function $Q_{(.,p)}$ is such that $Q_{(.,p)}(x_n) = Q_{(n,p)}$, and

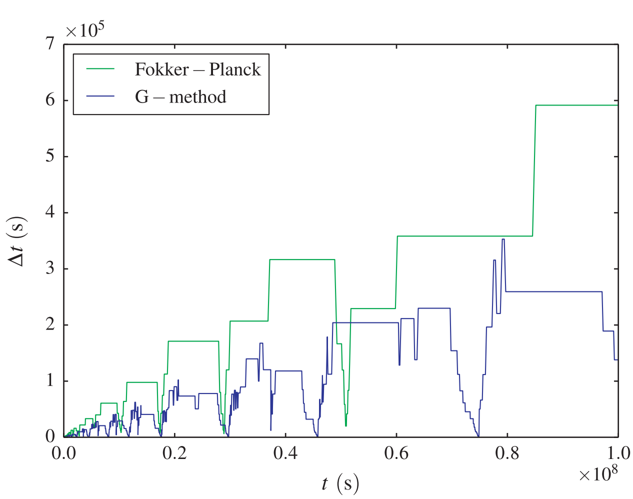

Fig. 9. Evolution of the time step Δt taken by the CVODES solver as a function of the physical time, when the Fokker–Planck scheme or the G-method are used on the test of Ref. [35].

$$R_{(.,p)}(x) = \sum_{m=-m_v}^{m_i}\sum_{q=0}^{\min(m_s,p)}\left[\beta_{((.,p-q),(m,q))}(x)C_{(.,p-q)}(x)C_{(m,q)} - \alpha_{((.,p),(m,q))}(x)C_{(.,p)}(x)\right] - \sum_{m=-m_v}^{m_i}\sum_{q=0}^{m_s}\left[\beta_{((.,p),(m,q))}(x)C_{(.,p)}(x)C_{(m,q)} - \alpha_{((.,p+q),(m,q))}(x)C_{(.,p+q)}(x)\right],$$

$$FC_{(.,p)}(x) = \sum_{m=-m_v}^{m_i}\sum_{q=0}^{\min(m_s,p)} m\beta_{((.,p-q),(m,q))}(x)C_{(.,p-q)}(x)C_{(m,q)} - \sum_{m=-m_v}^{m_i}\sum_{q=0}^{m_s} m\alpha_{((.,p+q),(m,q))}(x)C_{(.,p+q)}(x),$$

$$DC_{(.,p)}(x) = \sum_{m=-m_v}^{m_i}\sum_{q=0}^{\min(m_s,p)} \frac{1}{2}m^2\beta_{((.,p-q),(m,q))}(x)C_{(.,p-q)}(x)C_{(m,q)} + \sum_{m=-m_v}^{m_i}\sum_{q=0}^{m_s}\frac{1}{2}m^2\alpha_{((.,p+q),(m,q))}(x)C_{(.,p+q)}(x).$$

If clusters (m, q) containing self-defects and solute atoms are immobile, the absorption and emission rates of such clusters are zero. The term $R_{(.,p)}$ is then related to transitions between clusters due to the exchange of pure solute clusters, while FC and DC do not involve summations on q anymore. The same discretization scheme as presented above can be used.

If a Taylor expansion is also performed along the solute axis, the following equation is obtained:

$$\frac{\partial C}{\partial t} = -\frac{\partial J_x}{\partial x} - \frac{\partial J_y}{\partial y}, \tag{51}$$

where y is related to the solute axis,

$$J_x(x,y) = (F_xC)(x,y) - \frac{\partial D_{xx}C}{\partial x} - \frac{\partial D_{xy}C}{\partial y}, \tag{52}$$

$$J_y(x,y) = (F_yC)(x,y) - \frac{\partial D_{xy}C}{\partial x} - \frac{\partial D_{yy}C}{\partial y} \tag{53}$$

and

$$F_x(x,y) = \sum_{m=-m_v}^{m_i}\sum_{q=0}^{m_s} m\left[\beta_{(m,q)}(x,y)C_{(m,q)} - \alpha_{(m,q)}(x,y)\right], \tag{54}$$

$$F_y(x,y) = \sum_{m=-m_v}^{m_i}\sum_{q=0}^{m_s} q\left[\beta_{(m,q)}(x,y)C_{(m,q)} - \alpha_{(m,q)}(x,y)\right], \tag{55}$$

$$D_{xx}(x,y) = \frac{1}{2}\sum_{m=-m_v}^{m_i}\sum_{q=0}^{m_s} m^2\left[\beta_{(m,q)}(x,y)C_{(m,q)} + \alpha_{(m,q)}(x,y)\right], \tag{56}$$

$$D_{yy}(x,y) = \frac{1}{2}\sum_{m=-m_v}^{m_i}\sum_{q=0}^{m_s} q^2\left[\beta_{(m,q)}(x,y)C_{(m,q)} + \alpha_{(m,q)}(x,y)\right], \tag{57}$$

$$D_{xy}(x,y) = \frac{1}{2}\sum_{m=-m_v}^{m_i}\sum_{q=0}^{m_s} mq\left[\beta_{(m,q)}(x,y)C_{(m,q)} + \alpha_{(m,q)}(x,y)\right]. \tag{58}$$

As before, if clusters containing solutes and self-defects are immobile, expressions can be greatly simplified. In particular, the cross term D_{xy} is zero, so the scheme derived for the one dimensional Fokker–Planck equation can be used without modification.

In general, the cluster space can therefore be represented as in Fig. 10, where the various PDEs derived in the previous and in the present section are used. In CRESCENDO, formulas derived from a Taylor expansion along the two axes are implemented only in the case where no mobile clusters contain both self-defects and solute atoms. To test the Fokker–Planck approach in two dimensions, we consider a modified version of the test case exposed previously and detailed in Ref. [35], which consists in the aging of a population of quenched-in vacancies. To obtain a cluster distribution that expands both along the self-defect and solute axes, the initial condition is modified: the atomic fraction of quenched-in vacancies is 10^{-6} instead of 10^{-7} and we add a concentration of helium atoms equal to half the concentration of vacancies. Other new parameters are the diffusion prefactor and the migration energy for the diffusion of interstitial helium, which are chosen to be $D^0_{\mathrm{He}} = 8 \times 10^{-4}\ \mathrm{cm^2\ s^{-1}}$ and $E^{\mathrm{m}}_{\mathrm{He}} = 0.15$ eV, following Ref. [26]. Loop punching and interstitial emission by bubbles are not allowed, to avoid any complication due to the presence of self-interstitials. Binding energies of helium and vacancies are given by a bubble model described in Ref. [16]. Finally the formation of helium clusters is not permitted. Cluster distributions at $t = 10^8$ s are shown in Fig. 11 for the fully discrete calculation and for our Fokker–Planck approach, using the Chang–Cooper scheme. The discrete part corresponds to the region $[-10, 0] \times [0, 10]$ and outside this region, a geometrical progression with $\eta = 1.03$ is used. It can be seen that both distributions are very close in all regions of the cluster space.

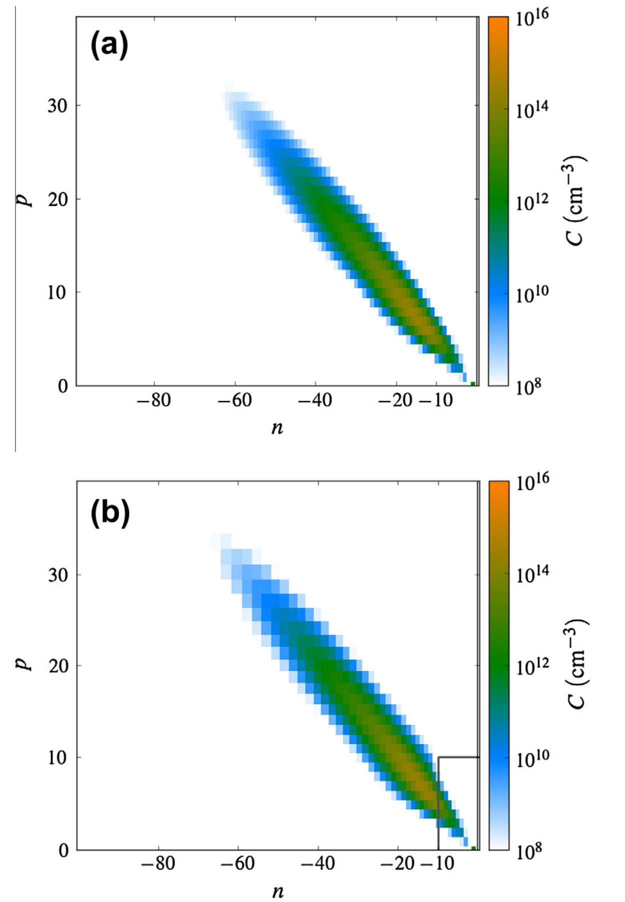

Fig. 11. Cluster distribution at $t = 10^8$ s using the modified parametrization of the test presented in Ref. [35], to include helium (see main text). (a) Fully discrete calculation. (b) Calculation using a Fokker–Planck approach with a geometrical progression equal to $\eta = 1.03$ along the vacancy and solute axes. The gray lines identify the discrete part.

4. Numerical solvers and performance issues

In this section we describe numerical aspects of the solving of the ODE system described in the previous section. To solve the ODE system, we use the CVODES library [37], which is well adapted in terms of numerical methods and which also provides several useful functionalities. For example, it is possible to add quadrature variables to the original ODE system, which can be used to compute integrated fluxes between all the elements of the microstructure at nearly no computational overhead. Another interesting feature is the ability of the code to compute the sensitivity of the results (cluster concentrations) with respect to input parameters (binding and migration energies, source term, etc.) in a single calculation. This permits to identify the key parameters for a given study.

We describe in the following two modifications that we have made to the original implementation of CVODES, which improve the solver efficiency for large problems. The first one is related to the linear solvers, while the second one permits to adapt on the fly the number of equations integrated with CVODES.

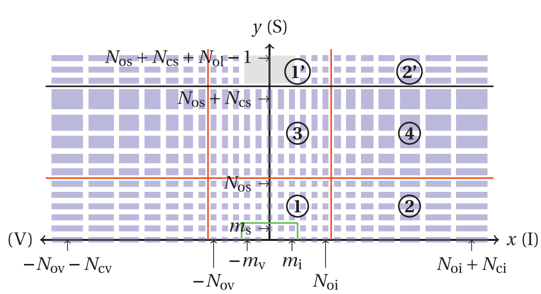

Fig. 10. Cluster space in a given slice when solute atoms and additional cluster populations are used. The transition part is omitted in this schematic. Each blue rectangle refers to a single equation, and its size reflects the mesh size in the cluster space. The number of equations in the discrete and the continuous parts along a given axis (v, i, or s) are indexed with 'o' and 'c' labels respectively, so that for example N_{ci} is the number of equations in the continuous part for interstitials. The number of different cluster populations is N_{ol}. In the different parts various equations are used. Parts 1 and 1′: Eqs. (6) and (15); parts 2, 2′: Eq. (50) (similar equation for part 3); part 4: Eq. (51). Parts 1′ and 2′ govern the evolution of additional cluster populations. The green line surrounds the part where clusters can be mobile. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4.1. Linear solvers

The ODE system (5) is numerically stiff, which roughly means that some modes have very small time constants with respect to others. In such a case, implicit integration methods are necessary. CVODES includes the Backward Differentiation Formulas (BDF), which are efficient for stiff problems. This implies the resolution of a linear system at each step. Standard linear solvers (direct dense, iterative) are provided by CVODES, as well as an easy way to link to alternative ones. We have coupled CVODES to two new linear solvers in order to speed up the computation and lower the memory requirement: MUMPS [38,39], which is currently one of the most powerful direct generic solver, and a solver based on Schur complement that exploits the specificities of our system.

The matrix of the linear system is a linear combination of the identity matrix and the jacobian matrix J of the ODE system, which is defined by

$$J_{ij} = \frac{\partial \dot{C}_i}{\partial C_j},$$

where $\dot{C}_i$ is the time derivative of C_i. This matrix is often sparse, as immobile clusters are coupled with each other only through mobile clusters, which are assumed to be just a few compared to the total number of cluster classes. MUMPS can handle any kind of sparsity, but the Schur solver exploits the specific sparse structure of the matrix, which we detail now.

When correctly reordered, the jacobian matrix can be written as

$$J = \begin{pmatrix} A & B \\ C & D \end{pmatrix},$$

where A is a square matrix which contains the couplings between immobile clusters, D is a square matrix which contains interactions between mobile clusters, B and C contain couplings between mobile and immobile clusters. When the number of mobile classes is small, A is a large sparse matrix and D is a small dense matrix. More precisely, A has the shape of a two-dimensional discretization scheme on a rectangular grid and thus appears as a band matrix. Therefore, the bandwidth roughly depends on the number of mobile clusters and, when both self-defects and solutes are introduced in the simulations, on the smallest size of the cluster space along self-defects or solutes. To minimize the bandwidth it is indeed interesting to make indices vary faster along self-defects or along solutes, depending on the shape of the cluster space, more precisely on dimensions $N_{ov} + N_{cv} + N_{oi} + N_{ci} + 1$ and $N_{os} + N_{cs} + 1$ (Fig. 10). The properties of matrices A and D make the use of the Schur complement particularly interesting to solve the linear system [40]. In addition, this method of solving can be parallelized by writing A as a block matrix. Alternatively, for spatialized calculations, it is interesting to partition the different slices on the different processors, so that blocks contain all immobile clusters of one slice or more.

As an example, we compare here the performances of the different linear solvers in the first case presented in Section 3.4 (Ref. [33]), with a final physical time equal to $t = 10^5$ s. No continuous part was used in this calculation, so 15,820 equations were necessary to simulate the cluster distribution. The calculation with the dense solver required 3.7 GB of memory and completed after around 14 h on a 3.07 Ghz processor. The use of MUMPS and Schur solvers led to a memory use of 44 MB and 27 MB, and to a CPU time of 108 s and 36 s respectively. In general, the Schur solver is all the more efficient as the bandwidth of the matrix A is small. It thus outperforms MUMPS typically for problems without solutes (which is the case here) or problems where the total number of solutes classes $N_{os} + N_{cs} + 1$ remains low compared to the total number of self-defect classes $N_{ov} + N_{cv} + N_{oi} + N_{ci} + 1$ (or vice versa, depending on the numbering of classes).

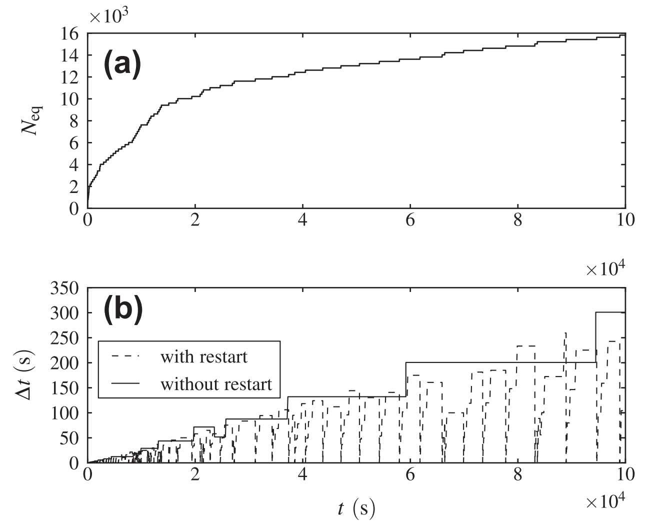

Fig. 12. (a) Variation of the number of equations when the cluster space resizing is used, as a function of the physical time, for the first test presented in Section 3.4. (b) Evolution of the time step Δt taken by the solver when it is restarted or when the integration history is preserved during cluster space resizing.

4.2. On the fly cluster space adaptation

Another way to improve the efficiency of the numerical solving is to reduce the number of equations N_{eq}, by ignoring cluster classes when concentrations are zero. Sharafat and Ghoniem proposed to use a growth law to determine a priori the position of the cluster distribution in the cluster space [28]. This law was used in a dynamic remeshing procedure. Very recently, Xu et al.

Table 2
Parameters used for the simulation of helium implantation in α-iron and the subsequent annealing of the sample. Migration and mixing energy of interstitial helium are given in Ref. [49]. It should be noted that dislocation loops are considered immobile in presence of helium, in agreement with previous calculations [5]. Given the temperature used for the annealing, loops are assumed to be [1 0 0] loops [50]. The binding energy of helium atoms to interstitial clusters is a fit of empirical potential calculations [51]. It is consistent with a strong trapping of helium by straight dislocations [52,53].

Symbol	Description	Value	Unit
V_{at}	Atomic volume	1.182×10^{-23}	cm^{-3}
m_i	Index of the largest mobile interstitial cluster	3	
m_v	Index of the largest mobile vacancy cluster	4	
m_b	Index of the largest mobile helium cluster	1	
D_v^0	Diffusion prefactor of clusters	8.2×10^{-3}	$cm^2\ s^{-1}$
$E^m_{(1,0)}, E^m_{(2,0)}, E^m_{(3,0)}$	Migration energy of interstitial clusters	0.34, 0.42, 0.43	eV
$E^m_{(-1,0)}, E^m_{(-2,0)}, E^m_{(-3,0)}, E^m_{(-4,0)}$	Migration energy of vacancy clusters	0.67, 0.62, 0.35, 0.48	eV
$E^m_{(0,1)}$	Migration energy of interstitial helium	0.06	eV
$E^f_{(1,0)}$	Formation energy of the mono-interstitial	3.77	eV
$E^f_{(-1,0)}$	Formation energy of the mono-vacancy	2.12	eV
$E^f_{(0,1)}$	Mixing energy of the interstitial helium	4.39	eV
$E^b_{(n,p),(1,0)}\ (n > 5)$	Binding energy of interstitials to interstitial clusters	Elastic law	
$E^b_{(n,p),(0,1)}\ (n > 0)$	Binding energy of helium to interstitial clusters	$0.242 + 0.058n$	eV
$b_\perp$	Component of the Burgers vector perpendicular to the loop	0.287	nm
$b_\parallel$	Component of the Burgers vector in the loop plane	0	
ρ_c	Cut-off parameter for the elastic law	$b_\perp/2 = 0.143$	nm
μ_s	Shear modulus	82	GPa
ν_p	Poisson's ratio	0.29	
$Z_{d,\mu}\ (\mu > 0)$	Absorption efficiency (straight dislocations)	1.1	
$Z_{d,\mu}\ (\mu < 0)$	Absorption efficiency (straight dislocations)	1.	
r_{iv}	Recombination radius	0.65	nm
ρ_d	Dislocation density (implantation/annealing)	10^8/ 0	cm^{-2}
F_d	Binding energy of helium to dislocations	∞	

proposed to use the values of the concentration and concentration gradient of clusters at the boundaries of the cluster space to determine if it should be enlarged [41]. Later, these authors used another method, requiring a calculation with a small cluster space to guess the cluster distribution on the whole space [42]. No dynamic space resizing was used in this case. As noticed by the authors, in general it is difficult to guess the evolution of the cluster distribution, so it is interesting to develop a method which permits to resize the cluster space on the fly without knowledge of its long time evolution, and without affecting too much the performances of the ODE solver.

For an irradiation problem, if the material has not been previously irradiated, the cluster space is minimal: it contains all mobile clusters and clusters that can be generated in displacement cascades. To determine if the cluster space should be enlarged in a given slice of the system, we check concentrations at the boundary of the space ($n = N_i$ or $n = -N_v$ or $p = N_s$) at each evaluation of the right hand side function of the ODE system. If they are larger than a given threshold at a given boundary, the boundary is shifted by a given amount ΔN_i, ΔN_v or ΔN_s.

In the standard use of CVODES, changing the number of equations requires the solver to be restarted. At each restart, the time step used by CVODES is initially very small and grows slowly, since the past history is lost. A restart can thus be very time consuming.

That is why we use another method to keep the efficiency of the multistep method. When a change in the cluster space is detected with the new concentration vector, the integration is stopped but the solver is not reinitialized. The internal data structure used by CVODES is reallocated with the appropriate size, keeping the existing values and extending with zeroes. This procedure allows us to keep the history and therefore induces no perturbations on the time step evolution. A similar approach can be used to decrease the size of the cluster space, which can be useful for annealing simulations.

The evolution of the time step as a function of the physical time is shown in Fig. 12 for the simulation used as a test case in the previous section, with an adjustable cluster space and $\Delta N_i = -\Delta N_v = 200$. Using the Schur solver, this leads to a CPU time equal to $t_{CPU} = 18$ s when a restart is performed and $t_{CPU} = 8$ s when the solver is not reinitialized. These values should be compared with the CPU time $t_{CPU} = 36$ s obtained when a fixed cluster space with 15,820 equations is used. The first thing to note is that even if the solver is reinitialized, it is preferable to use a variable cluster space. In addition, as shown in Fig. 12, a substantial gain in computation time is obtain when the history is kept at each resizing, since it avoids the drop in time steps observed at each resizing. Such a method is all the more important for spatialized simulations, for which the solver is restarted every time a space resizing occurs in a particular slice.

Table 3
Total binding energies of vacancy clusters used for the simulation of helium implantation in α-iron and the subsequent annealings [54]. Total binding energies are defined as the difference between the formation energy of the fully dissociated cluster and the formation energy of the cluster. For bubbles with $n \leqslant -5$, the bubble model proposed in Ref. [16] is used.

n	p	$E^{b,tot}_{(n,p)}$ (eV)	n	p	$E^{b,tot}_{(n,p)}$ (eV)
0	2	0.43	−3	0	0.67
0	3	1.38	−3	1	3.90
0	4	2.37	−3	2	6.78
0	5	3.59	−3	3	9.62
−1	1	2.30	−3	4	12.19
−1	2	4.14	−3	5	14.51
−1	3	5.97	−4	0	1.29
−1	4	7.85	−4	1	5.06
−2	0	0.30	−4	2	8.10
−2	1	3.08	−4	3	11.19
−2	2	5.75	−4	4	14.17
−2	3	7.82	−5	0	2.01
−2	4	10.18	$n \leqslant -5$	$p \geqslant 1$	Model

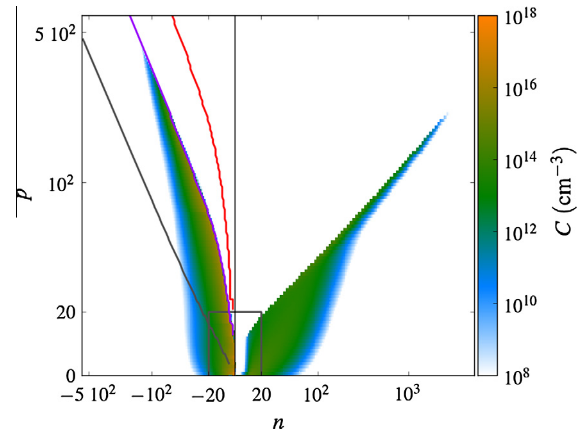

Fig. 13. Cluster distribution at $z = 180$ nm after implantation of 60 keV helium in α-iron at a fluence equal to 10^{16} cm^{-2}. The gray square identifies the discrete part. The scale is given by indexes n and p instead of x and y, so outside the discrete part the scale evolves geometrically. The equilibrium line for vacancies is draw in gray. The purple and red lines are the loci where the net flux of interstitials and helium atoms to clusters are zero.

5. Application: coarsening of bubbles

To illustrate the use of CRESCENDO for the simulation of large scale problems, we consider here the isothermal annealing of a sample irradiated at 300 K with 60 keV helium ions at a fluence equal to 10^{16} cm^{-2}, with a flux equal to 5.5×10^{12} He cm^{-2} s^{-1}. Such implantations are commonly used to probe the properties of helium bubbles by helium thermodesorption [43–45]. Thermodesorption cannot give direct evidence of what occurs in the material. To better understand the physical mechanisms, it can be coupled to transmission electron microscopy (TEM) [11]. However thermodesorption is often carried out on a thick sample whose depth is sufficient to avoid an effect of the back surface on the microstructural evolution, whereas surface effects can be large with TEM. It is therefore a priori difficult to correlate thermodesorption with TEM. Simulation provides a way to understand

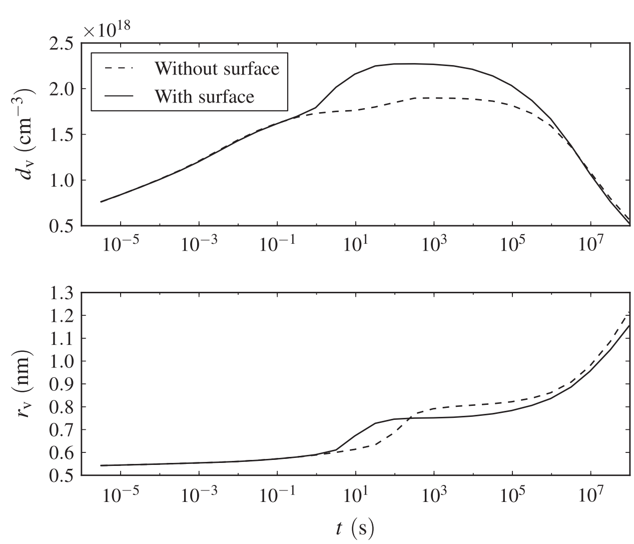

Fig. 14. Cluster density and mean radius for an isothermal annealing at 1073 K, at $z = 17$ nm in a 100 nm thick foil and in a system without surfaces.

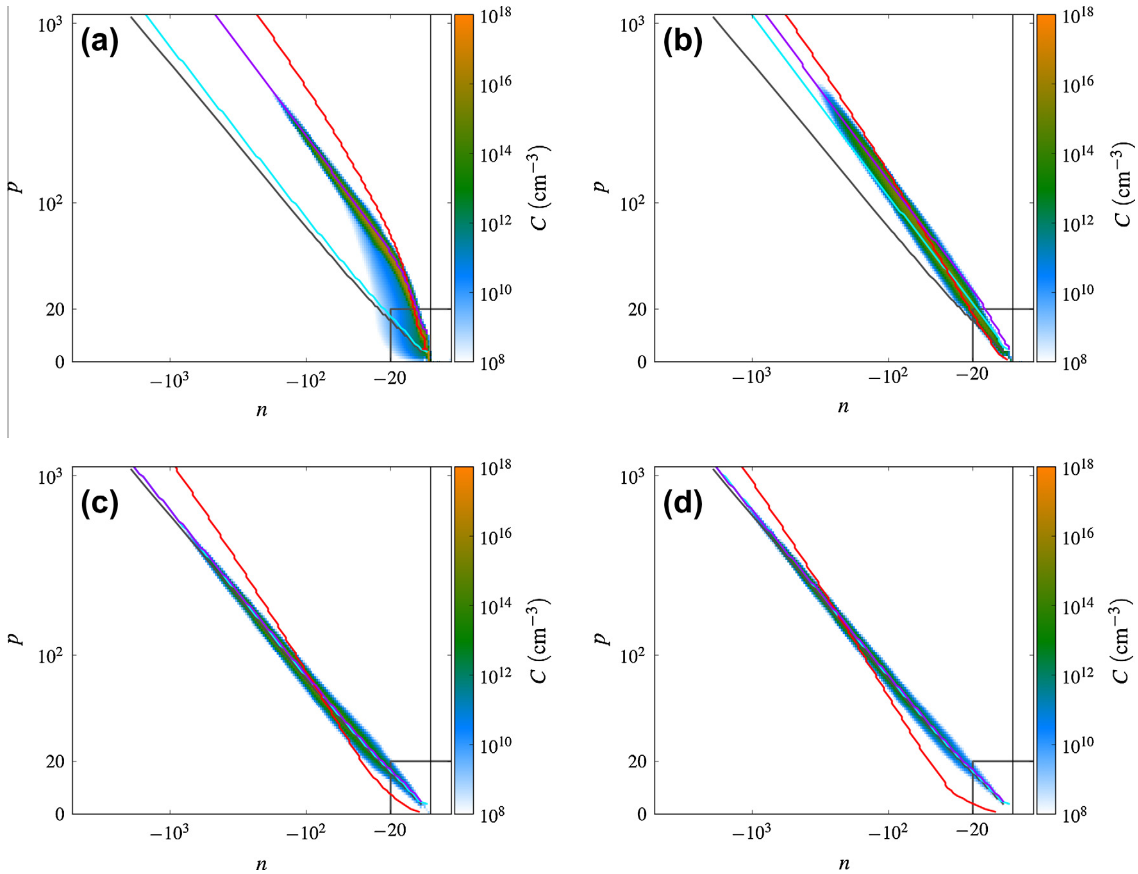

Fig. 15. Cluster distribution (vacancy clusters only) during an annealing at 1073 K, without surface, at different times: (a) 10^{-5} s, (b) 30 s, (c) 3000 s, and (d) 10^7 s. The gray line is the equilibrium line for vacancies. The cyan, purple and red lines are the loci where the net flux of vacancies, interstitials and helium atoms to clusters are zero. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

the effect of the sample's geometry on the microstructure. That is why after simulating the helium implantation we propose to model the annealing of a bulk sample and of a thin foil at 1073 K to highlight the role of surfaces. We focus more precisely on the evolution of helium bubbles. Although at this temperature bubbles have been shown to be mobile [46], we ignore this phenomenon to see the influence of surface on the coarsening by Ostwald ripening.

Helium implantation is simulated with CRESCENDO, using an effective source term based on the spatial homogenization of cascades [47] produced by MARLOWE [48], which ensures a correct reproduction of spatial correlations within displacement cascades. Parameters used for the calculations are given in Tables 2,3. A variable spatial mesh size is chosen with $N_{\text{grid}} = 20$ and a denser mesh close to the irradiated surface is used, in order to simulate precisely the irradiated area and to consider a sufficiently large system. The thickness of the system is 10 μm, and the helium outward flux at the back side is less than 1% of the helium flux at the front side at the end of implantation, so that the effect of the back surface can be safely neglected. It should be noted that slices at the back are only populated by small clusters and since the cluster space is adapted, more slices could be added without increasing too much the computation time.

After implantation, in order to mimic the TEM configuration, the cluster distribution close to the surface, at $z = 180$ nm, is used as an input for next calculations. The annealing is performed on a sample containing this cluster distribution over a thickness equal to 100 nm, which is the order of magnitude of the local thickness of samples observed by TEM. Surfaces are assumed to act as perfect sinks and sources for defects. To investigate surface effects, annealing is also performed with the same cluster distribution but with no surfaces. The initial cluster distribution is shown in Fig. 13. It exhibits small bubbles and small dislocation loops of interstitial type. The amount of helium is so large that loops, which are assumed to contain at most as many helium atoms as possible atomic sites on their dislocation line, are full of helium. This could possibly indicate that small vacancy–helium clusters are nucleated on the dislocation line, which has been seen in other experimental conditions [55]. Since heterogeneous nucleation is not present in the model, we suppose that if the number of helium atoms on the dislocation outstrips the maximum value, it is not bound to the loop. Concerning bubbles, to analyze the distribution we plot the equilibrium line for vacancies, *i.e.* the line where the chemical potential of vacancies in the matrix near bubbles is zero [56], so that no Gibbs–Thomson effect is present (the presence of helium counterbalances the curvature effect). It can be noted that bubbles do not lie on the equilibrium line for vacancies: most of them are overpressurized. On this graph the loci where the net fluxes of defects to clusters are zero are also shown [57,58], for the following defects: helium atoms (in red[2]), self-interstitials (in purple), and vacancies (in cyan, not visible on this graph). These net fluxes are computed as $\beta_{v,\mu}C_\mu - \alpha_{v,\mu}$. The fact that the bubble distribution is much populated below and not populated just above the interstitial line indicates that the emission of interstitials by bubbles is very high above this line. This emission mechanism is analogous to loop

[2] For interpretation of color in Fig. 13, the reader is referred to the web version of this article.

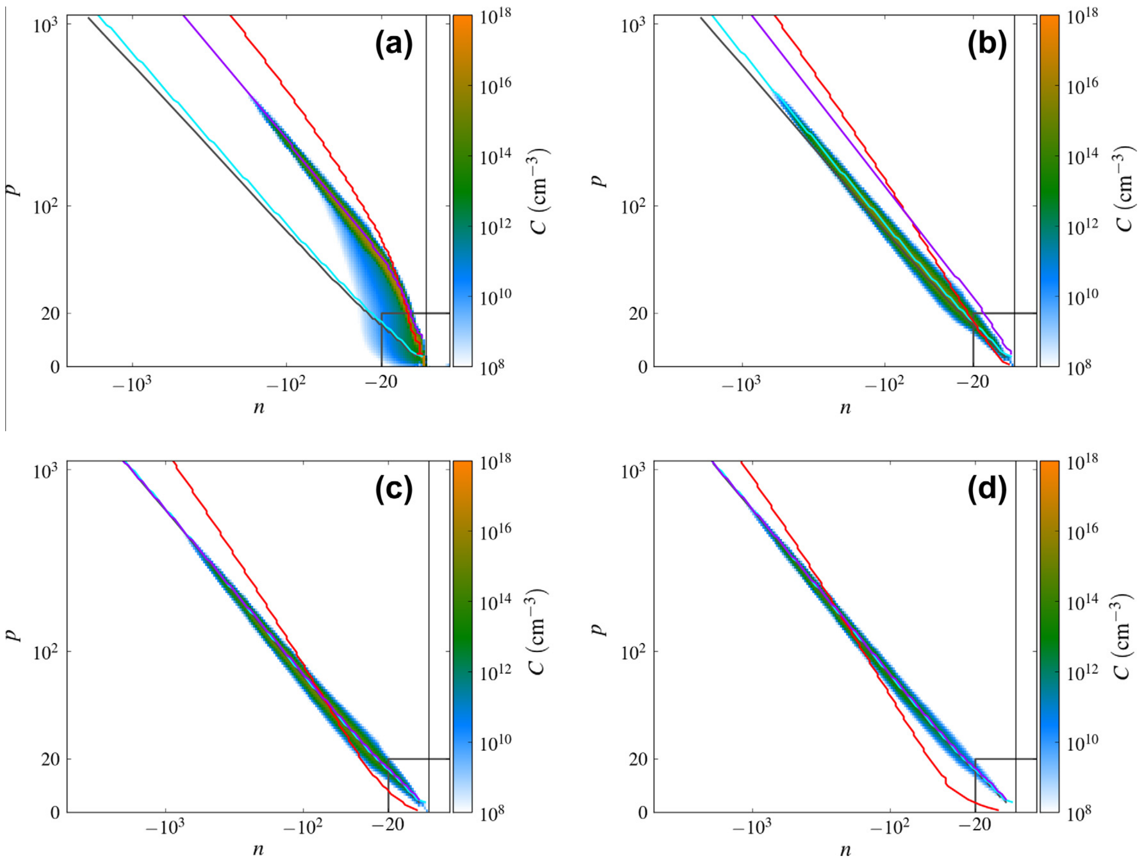

Fig. 16. Cluster distribution (vacancy clusters only) during an annealing at 1073 K of a 100 nm system, at $z = 17$ nm and at different times: (a) 10^{-5} s, (b) 30 s, (c) 3000 s, (d) 10^7 s. The cyan, purple and red lines are the loci where the net flux of vacancies, interstitials and helium atoms to clusters are zero. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

punching [56,59] and has been observed in molecular dynamics simulations [60].

The evolution of density and mean radius of clusters during the isothermal annealing at 1073 K is shown in Fig. 14 for the two kinds of systems. The minimum radius of a bubble taken into account in the calculation of the density is $r_v = 0.5$ nm. To analyze these curves, cluster distributions are plotted in Figs. 15 and 16 at different times of the annealing, only for vacancy clusters.

Just after implantation, the behaviors of the two systems are very similar. Compared to Fig. 13, the cluster distribution becomes more narrow and centered on the line corresponding to zero net flux of self-interstitials (Fig. 15(a) and 16(a)). This means that during this early evolution, interstitials have a large influence on the kinetics of the system. They are emitted mostly by overpressurized bubbles, but also by small interstitial loops, and captured by less pressurized clusters, which leads to the narrowing of the distribution. For small clusters, helium resolution and absorption also plays a role.

From a few tenths of second to a few seconds, the curves corresponding to the two experimental conditions depart from each other. A clear increase of the density and radius is observed in presence of a surface, whereas the cluster distribution in the bulk remains unchanged. In Figs. 15(b) and 16(b) it is clearly seen that the cluster distribution in the bulk remains on the interstitial line, whereas the cluster distribution close to a surface is now aligned on the vacancy line and nearly reaches the equilibrium line for vacancies. This can be justified by the fact that at this time, vacancies have been sufficiently generated at the surface and have diffused to the clusters to impose an equilibrium. Such a state is seen only later in the bulk, since vacancies can only come from clusters.

It seems that after around 10^3 s, both systems have reached an equilibrium. However the helium line is not aligned on the distribution, which means that the distribution still evolves. This slow evolution due to helium resolution and absorption is only seen over very large time scales, leading to the coarsening mechanism observed in Fig. 14 [61]. It should be noted that along the complete evolution process, the quantity of helium lost at surfaces is rather small, probably due to the high density of traps inside the matrix: at $t = 10^7$ s, only 5% of the initial quantity of helium is lost. The main role of surfaces is thus here to provide vacancies, and not to act as a sink for helium.

6. Conclusion

In this article we have presented the CRESCENDO code and described some general methods that can be included in cluster dynamics codes to simulate large scale problems under irradiation or under thermal aging. In particular, we have focused on a numerical scheme to solve the Fokker–Planck equation when it is coupled to standard rate equations, which strictly conserves matter. Although it describes the cluster distribution less precisely than the grouping method by Golubov et al. when the distribution is peaked, the loss of accuracy is moderate on the tests we performed and the computation time is greatly reduced, by a factor of 30 for the largest physical times. We have also discussed numerical methods that can be used to increase the efficiency of standard cluster dynamics simulations. The use of state-of-the-art sparse linear solvers, together with a dynamic remeshing of cluster size space without reinitialization of the ODE solver, are particularly efficient to handle large systems over realistic time scales.

As a example, we have simulated the implantation of helium in α-iron at a fluence of $10^{16}\,\mathrm{cm}^{-2}$, followed by the annealing at 1073 K. We have compared the annealing of a bulk sample to the annealing of a thin foil, as used in TEM experiments. The differences in the evolution of the bubble radius and density have been explained on the basis of the cluster size distributions.

Acknowledgments

CRESCENDO is co-developed by CEA and EDF R&D in the framework of the SINERGY program. The authors are grateful to A. Barbu (CEA) for his advice about CRESCENDO and to V. Duwig and M. Barrault (EDF R&D) for their initial contribution to the code.

The interface to the Schur solver in CVODES was originally developed by D. Guibert [62] in the framework of the PARADE project (ANR-06-TLOG-0026), and adapted to our case during the ParMat project (ANR-06-CIS6-006). The authors thank D. Tromeur-Dervout for giving us the permission to use the Schur solver and D. Guibert and E. Cancès for fruitful discussions.

Appendix A. Rate equations for mobile clusters

Equations for mobile clusters when Fokker–Planck equation is used have been sketched in Section 3.3. We derive here the full expression, which notably takes into account the transition part.

We consider for example the case of the continuous interstitial part ($n > 0$). We note i_1 the first index in the transition part. Since the numerical scheme is centered in the transition part and $\Delta x = 1$, the contribution of a mobile cluster $m > 0$ to the variation of matter reads

$$\begin{aligned}\frac{\mathrm{d}\mathcal{S}^m}{\mathrm{d}t} &= \sum_{n=i_1}^{\infty} x_n\left[J^m_{n-1/2} - J^m_{n+1/2}\right] \\ &= \sum_{n=i_1}^{i_1+m-1} x_n\left[\frac{m(m+1)}{2}\beta_{n-1,m}C_{n-1}C_m + (1-m^2)\beta_{n,m}C_nC_m + \frac{m(m-1)}{2}\beta_{n+1,m}C_{n+1}C_m\right] \\ &+ \sum_{n=i_1}^{i_1+m-1} x_n\left[\frac{m(m-1)}{2}\alpha_{n-1,m}C_{n-1} + (1-m^2)\alpha_{n,m}C_n + \frac{m(m+1)}{2}\alpha_{n+1,m}C_{n+1}\right] \\ &- \sum_{n=i_1}^{i_1+m-1} x_n\beta_{n,m}C_nC_m - \sum_{n=i_1}^{i_1+m-1} x_n\alpha_{n,m}C_n + \sum_{n=i_1+m}^{\infty} x_n\left(J^m_{n-1/2} - J^m_{n+1/2}\right).\end{aligned} \tag{A.1}$$

As discussed in Section 3.2, the first and the fourth summations introduce a coupling between the discrete and the transition part, and are therefore replaced by their exact expression if necessary. In the present case, the emission term α is exactly reproduced, so the change should only be done for the absorption term β. The second and third summations introduce a coupling between the transition and the continuous parts and are not altered.

It can be checked that if $\Delta x = 1$, if a centered scheme is used for the fluxes and the exact representation of these fluxes is taken, the contributions of the second, third and last term of Eq. (A.1) give an absorption flux equal to

$$\sum_{n=i_1}^{\infty} m\beta_{n,m}C_nC_m$$

and an emission flux equal to

$$-\sum_{n=i_1}^{\infty} m\alpha_{n+m,m}C_{n+m}.$$

As stated in Section 3.3, to obtain an equation for the mobile clusters m due to the internal fluxes of the transition and continuous parts, one must therefore divide these three contributions by m.

It should be noted that in the general case, the last term in Eq. (A.1) can also be written

$$\sum_{n=i_1+m}^{\infty} x_n\left(J^m_{n-1/2} - J^m_{n+1/2}\right) = (i_1+m)J^m_{i_1+m-1/2} + \sum_{n=i_1+m+1}^{\infty} \Delta x_{n-1}J^m_{n-1/2}. \tag{A.2}$$

We note $J^m_{n+1/2,\alpha}$ and $J^m_{n+1/2,\beta}$ the components of $J^m_{n+1/2}$ due to the emission and absorption terms respectively, without including C_m in $J^m_{n+1/2,\beta}$. We then define the following partial fluxes J^m_β and J^m_α, which contain the internal fluxes and the exact representations of the fluxes which cross the boundary between the discrete and transition parts:

$$J^m_\beta = \sum_{n=i_1}^{i_1+m-1}\beta_{n-m,m}C_{n-m} - \frac{1}{m}\sum_{n=i_1}^{i_1+m-1} n\beta_{n,m}C_n + \frac{i_1+m}{m}J^m_{i_1+m-1/2,\beta} + \frac{1}{m}\sum_{n=i_1+m+1}^{\infty}\Delta x_{n-1}J^m_{n-1/2,\beta} \tag{A.3}$$

and

$$\begin{aligned}J^m_\alpha = &-\sum_{n=i_1}^{i_1+m-1}\alpha_{n,m}C_n + \frac{1}{m}\left\{\sum_{n=i_1}^{i_1+m-1} n\left[\frac{m(m-1)}{2}\alpha_{n-1,m}C_{n-1} + (1 - m^2)\alpha_{n,m}C_n + \frac{m(m+1)}{2}\alpha_{n+1,m}C_{n+1}\right]\right\} \\ &+ \frac{i_1+m}{m}J^m_{i_1+m-1/2,\alpha} + \frac{1}{m}\sum_{n=i_1+m+1}^{\infty}\Delta x_{n-1}J^m_{n-1/2,\alpha}.\end{aligned} \tag{A.4}$$

In addition to the contribution of fluxes between classes in the discrete part, the following two terms are added to account for the contribution of the transition and continuous parts:

$$\frac{\mathrm{d}C_m}{\mathrm{d}t} = \cdots - J^m_\beta C_m - J^m_\alpha. \tag{A.5}$$

Similar expressions to Eqs. (A.3) and (A.4) can be found for $m < 0$ and for the vacancy part ($n < 0$).

Appendix B. Numerical diffusion of the numerical scheme for the Fokker–Planck equation

In this section we estimate the amount of numerical diffusion introduced by the discretization scheme described in Section 3.1. Using Eq. (37), the right hand side of the Fokker–Planck equation reads

$$\frac{2(J_{n-1/2} - J_{n+1/2})}{\Delta x_{n-1} + \Delta x_n} = \left[(1-\delta_{n-1})(FC)_n - \frac{(DC)_n}{\Delta x_{n-1}} + \delta_{n-1}(FC)_{n-1} + \frac{(DC)_{n-1}}{\Delta x_{n-1}} - (1-\delta_n)(FC)_{n+1} + \frac{(DC)_{n+1}}{\Delta x_n} - \delta_n(FC)_n - \frac{(DC)_n}{\Delta x_n}\right]\frac{2}{\Delta x_{n-1} + \Delta x_n}. \tag{B.1}$$

Using a second-order Taylor series expansion of $(FC)_{n-1}$, $(FC)_{n+1}$, $(DC)_{n-1}$ and $(DC)_{n+1}$, this equation is consistent with

$$\frac{\partial C}{\partial t} = -\frac{2(\Delta x_{n-1}\delta_{n-1} + (1-\delta_n)\Delta x_n)}{\Delta x_{n-1} + \Delta x_n}\frac{\partial FC}{\partial x} + \frac{\delta_{n-1}\Delta x^2_{n-1} - (1-\delta_n)\Delta x^2_n}{\Delta x_{n-1} + \Delta x_n}\frac{\partial^2 FC}{\partial x^2} + \frac{\partial^2 DC}{\partial x^2}. \tag{B.2}$$

If we assume that $DC = \gamma FC$, we see that the discretization scheme introduces an additional diffusion term, so that the diffusion term should be multiplied by

$$f_n = 1 + \frac{\delta_{n-1}\Delta x^2_{n-1} - (1-\delta_n)\Delta x^2_n}{\Delta x_{n-1} + \Delta x_n}\frac{1}{\gamma}. \tag{B.3}$$

It is easily checked that a centered scheme ($\delta_n = 1/2$) on a uniform mesh drastically reduces the numerical diffusion.

In the specific case where a geometric progression is used ($\Delta x_n = \eta\Delta x_{n-1}$), Eq. (B.2) becomes

$$\frac{\partial C}{\partial t} = -\frac{2[\delta_{n-1} + (1-\delta_n)\eta]}{1+\eta}\frac{\partial FC}{\partial x} + \Delta x_{n-1}\frac{\delta_{n-1} - (1-\delta_n)\eta^2}{1+\eta} \times \frac{\partial^2 FC}{\partial x^2} + \frac{\partial^2 DC}{\partial x^2}. \tag{B.4}$$

It is interesting to perform the same type of analysis with the Kiritani method in the case of the capture and emission of a single type of monomer ($m = 1$), to understand the differences between the two schemes in the Ostwald ripening test performed in Section 3.4. We obtain

$$\frac{\partial C}{\partial t} = -\frac{2}{1+\eta}\frac{\partial F'C}{\partial x} + \Delta x_{n-1}\frac{2}{1+\eta}\frac{\partial^2 D'C}{\partial x^2}, \tag{B.5}$$

with

$$F' = \beta C_1 - \eta^2\alpha \tag{B.6}$$

$$D' = \frac{1}{2}(\beta C_1 + \eta^3\alpha). \tag{B.7}$$

Two effects can thus explain the shift to smaller sizes for the Kiritani method. As for the Fokker–Planck method discretized with an upwind scheme ($\delta_{n-1} = \delta_n = 1$), it introduces a factor $2/(1+\eta)$, which slightly reduces the advection term. In addition, by comparing F' to $F = \beta C_1 - \alpha$, it is seen that this method introduces an effect of group size in the balance between absorption and emission, which induces the additional shift seen in the inset of Fig. 7.

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